# Math 420 - Exercise Set 3

**1.** Let $(X, \mathcal{T})$ be a topological space, $A$ and $B$ be subsets of $X$. Prove the following :

**(a)** $A$ is closed $\Leftrightarrow A = \bar{A}$

**(b)** $\bar{A} = A^\circ \cup \partial A$

**(c)** $\bar{A}$ is the smallest closed subset of $X$ containing $A$

**(d)** $A^\circ = X \setminus (\overline{X \setminus A})$

**(e)** $ext(A) = X \setminus \bar{A}$

**(f)** $\partial A = \bar{A} \cap (\overline{X \setminus A})$

**(g)** $A^\circ \subseteq (\bar{A})^\circ$

**(h)** $ext(A) = ext(\bar{A})$

**(i)** $\partial \bar{A} \subseteq \partial A$

**(j)** $\overline{(\overline{A})} = \overline{A}$

**(k)** $\overline{A \cap B} \subseteq \bar{A} \cap \bar{B}$

**(l)** $\overline{A \cup B} = \bar{A} \cup \bar{B}$

**(m)** $\partial A^\circ \subseteq \partial A$.

**(n)** How does $\partial(A \cup B)$ relate to $\partial A \cup \partial B$?

**(o)** $\partial A = \partial(X \setminus A)$

**(p)** If $A \subseteq B$ then $\overline{A} \subseteq \overline{B}$

**(q)** How does $\partial(ext(A))$ relate to $\partial(A)$?

**2.** Let $(X, \mathcal{T})$ be a topological space, $U \in \mathcal{T}$ and $A$ be any subset of $X$. Show that $U \cap \overline{A} \subseteq \overline{U \cap A}$.

**3.** Let $(X, \mathcal{T})$ be a topological space and $A$ a dense subset of $X$. If $U \in \mathcal{T}$ show that $U \subseteq \overline{(A \cap U)}$

**4.** Let $(X, \mathcal{T})$ be any topological space and $(\mathbb{R}, \mathcal{U})$ be the usual topological space. If $f : X \to \mathbb{R}$ is a continuous function determine if the set $A = \{x \in X \mid f(x) = 0\}$ is a closed set.

**5.** Let $X = \{a, b, c, d\}$ and let $\mathcal{T} = \{U \mid U \subseteq X \text{ and } c \notin U\} \cup \{X\}$ be the excluded point topology on $X$.

**(a)** Write all $\mathcal{T}$-open subsets of $X$.

**(b)** Write all $\mathcal{T}$-closed subsets of $X$.

**(c)** If $A = \{a, c, d\}$ then determine the following subsets of $X$ : $A^\circ$, $ext(A)$, $\partial A$, $\bar{A}$, $\partial(\bar{A})$, $I(A)$ and $\tilde{A}$.

**6.** Let $(X, \mathcal{T})$ be a topological space and $K$ a $\mathcal{T}$-closed subset of $X$. If $X = A \cup K$ for every subset $A$ of $X$ then show that $X = A^\circ \cup K$.

**7.** If $f : (X, \mathcal{T}) \to (Y, \mathcal{V})$ is a $\mathcal{T} - \mathcal{V}$-continuous function show that $\overline{f^{-1}(B)} \subseteq f^{-1}(\overline{B})$ for any subset $B$ of $Y$.

**8.** Let $X$ be a set, $x_0 \in X$ a fixed point and

$$\mathcal{T} = \{U \mid U \subseteq X \text{ and } x_0 \in U\} \cup \{\emptyset\}.$$

**(a)** Show that $\mathcal{T}$ is a topology on $X$. ( $\mathcal{T}$ is called the included point topology or the particular point topology on $X$.)

**(b)** Let $X = \mathbb{Q}$ and $\mathcal{T}$ be the includede point topology on $\mathbb{Q}$ with $x_0 = 1$. If $A = \mathbb{Z}$ find $A^\circ$, $ext(A)$, $\overline{A}$, the set $I(A)$ of isolated points and the derived set $\tilde{A}$ of $A$.

**(c)** Determine if $A$ is a dense subset of $\mathbb{Q}$.

**9.** Let $X$ be a nonempty set, $x_0 \in X$ be a point. If $\mathcal{T} = \{\emptyset, X\} \cup \{U \mid U \subseteq X,\ x_0 \in U\}$.

**(a)** Show that $\mathcal{T}$ is a topology on $X$. It is called the particular point topology on $X$.

**(b)** Find all $\mathcal{T}$-closed subsets of $X$.

**(c)** If $(x_n)$ is a sequence of points in $X$ converging to $x_0$ what can we say about the sequence $(x_n)$?

**(d)** Consider the particular point topology on $\mathbb{R}$ with $x_0 = 5$. If $A = \{3, 6\}$ find the interior, exterior, boundary, closure, set of isolated points $I(A)$ and the derived set $\tilde{A}$ of $A$.

**10.** Determine which of the following classes of subsets of $\mathbb{R}$ are bases :

**(a)** $\mathcal{B}_1 = \{(n, n+2) \subset \mathbb{R} \mid n \in \mathbb{Z}\}$

**(b)** $\mathcal{B}_2 = \{[a, b] \subset \mathbb{R} \mid a, b \in \mathbb{R},\ a < b\}$.

**11.** Let $X = \mathbb{R}^2$ and $\mathcal{B} = \{\{a\} \times (b, c) \subset \mathbb{R}^2 \mid a, b, c \in \mathbb{R}\}$ be the collection of vertical intervals in the plane. Show that $\mathcal{B}$ is a base for a topology on $\mathbb{R}^2$. This topology is called the vertical interval topology.